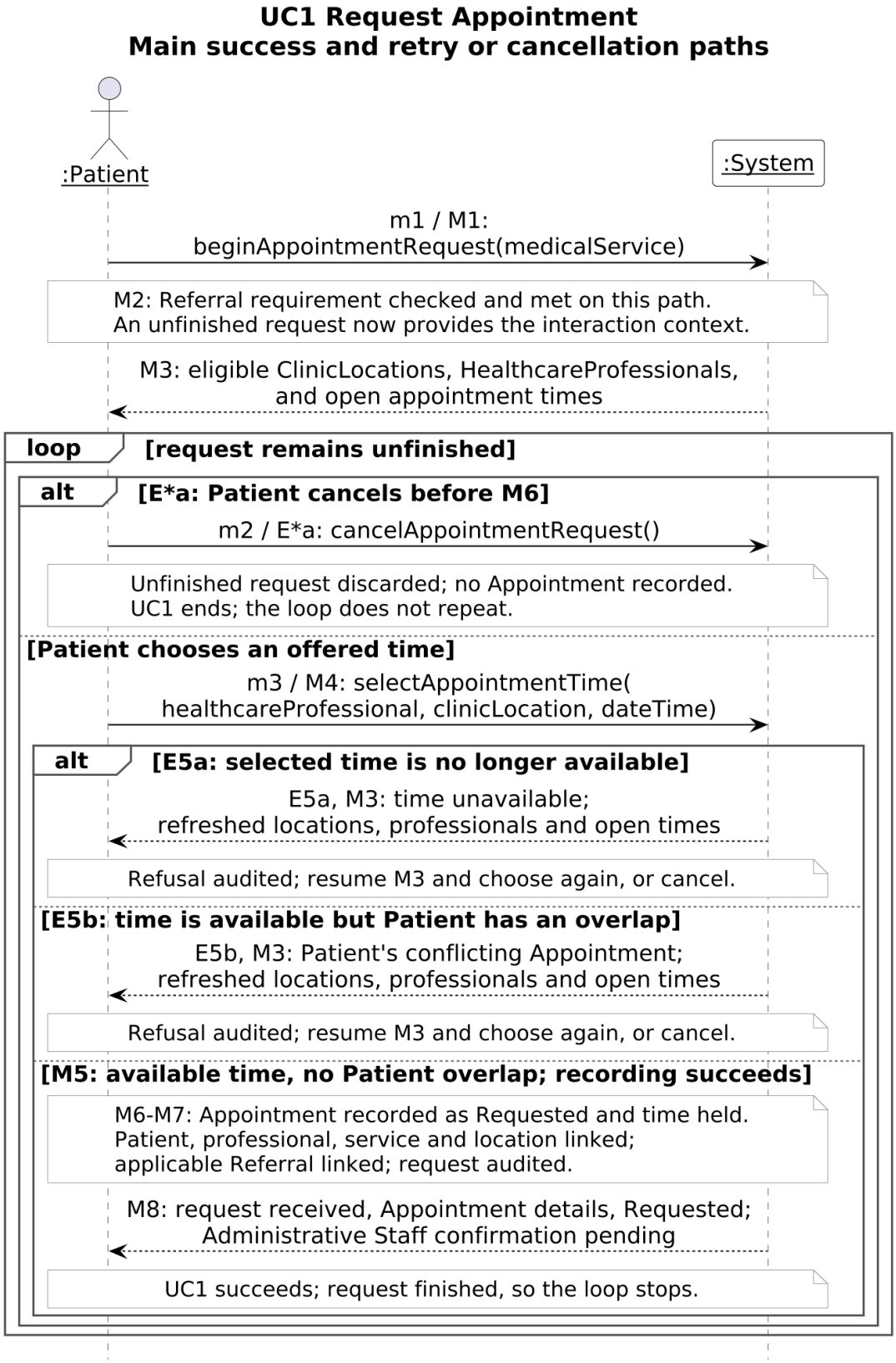
System sequence diagrams — UC1-S

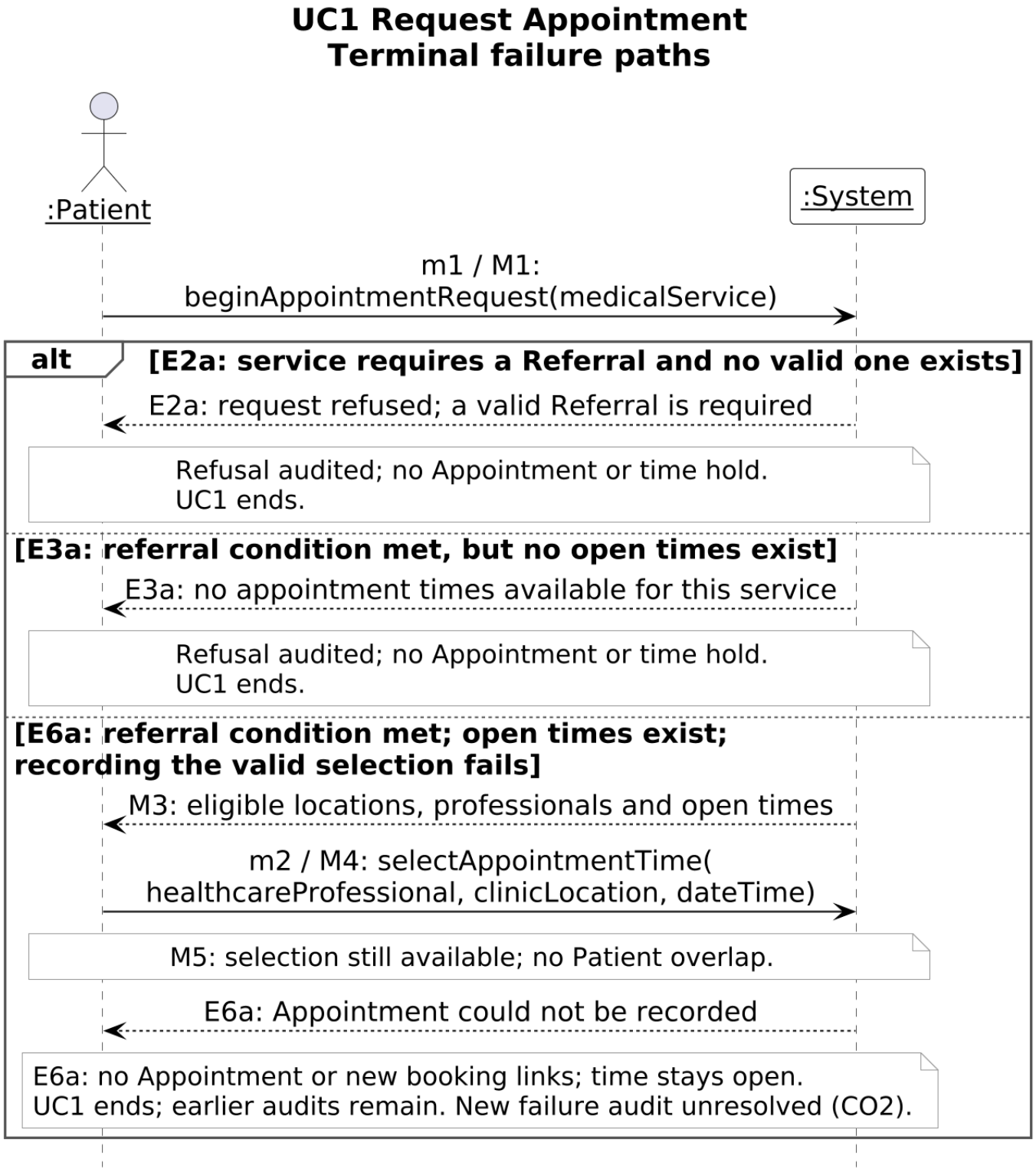
Main steps 1-8; extensions 5a, 5b and *a. The main success path takes the valid-selection branch on the first pass. A registered, identified Patient supplies the MedicalService and appointment choice.

M = main step; E = extension; m = local incoming message. Sources: UC1-UC3 and p01.iter1.pdf. Method: L7 pp. 2-7; L8 pp. 2-10; L9 p. 7; L10 pp. 6-7, 15-17; L11 p. 8. Each diagram treats the Appointment Tracking System as one black box.



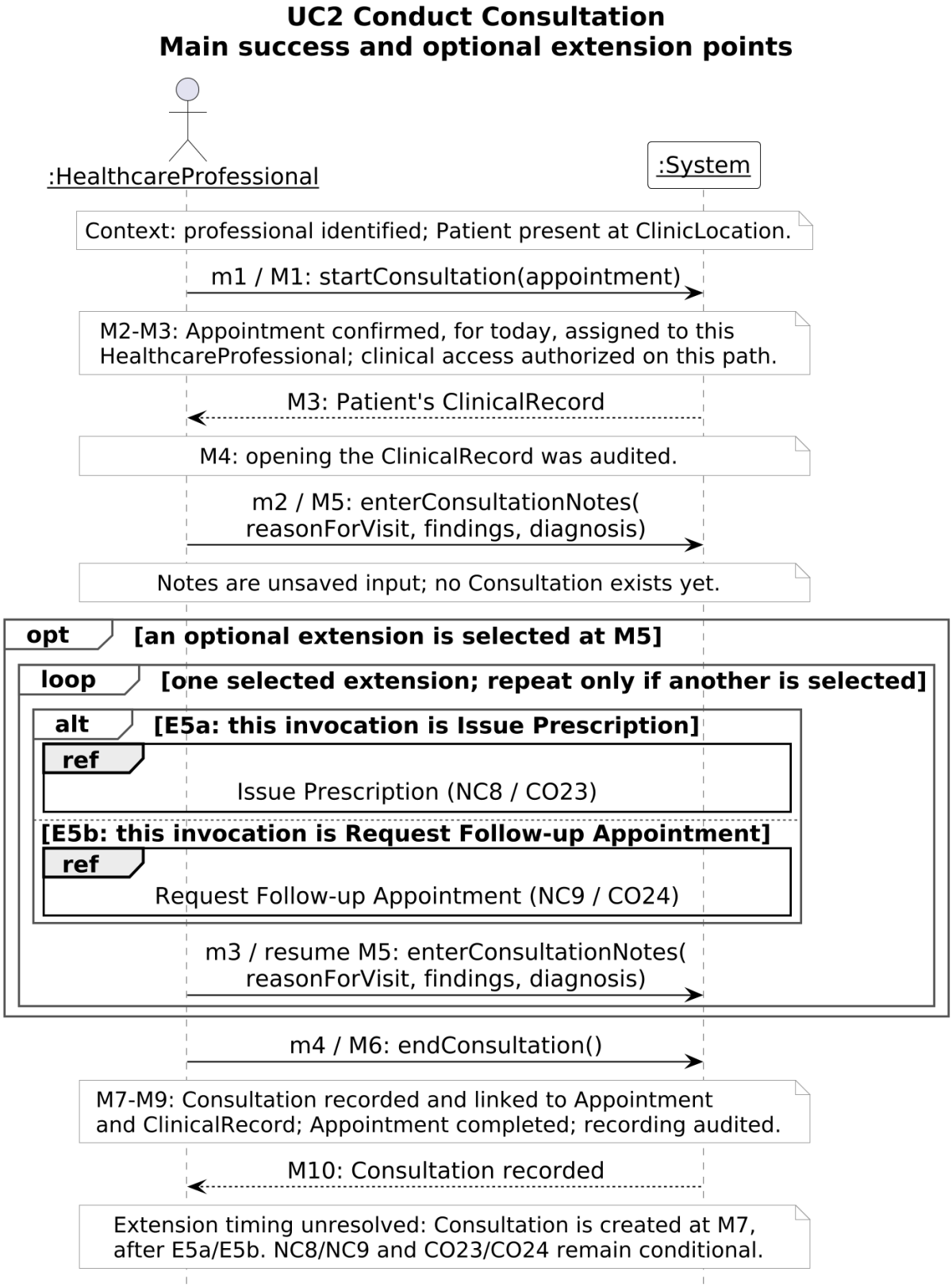
System sequence diagrams — UC1-F

Extensions 2a, 3a and 6a. Every branch terminates without recording an Appointment. The save-failure branch follows a valid selection; it is distinct from a time-selection rejection that allows retry.



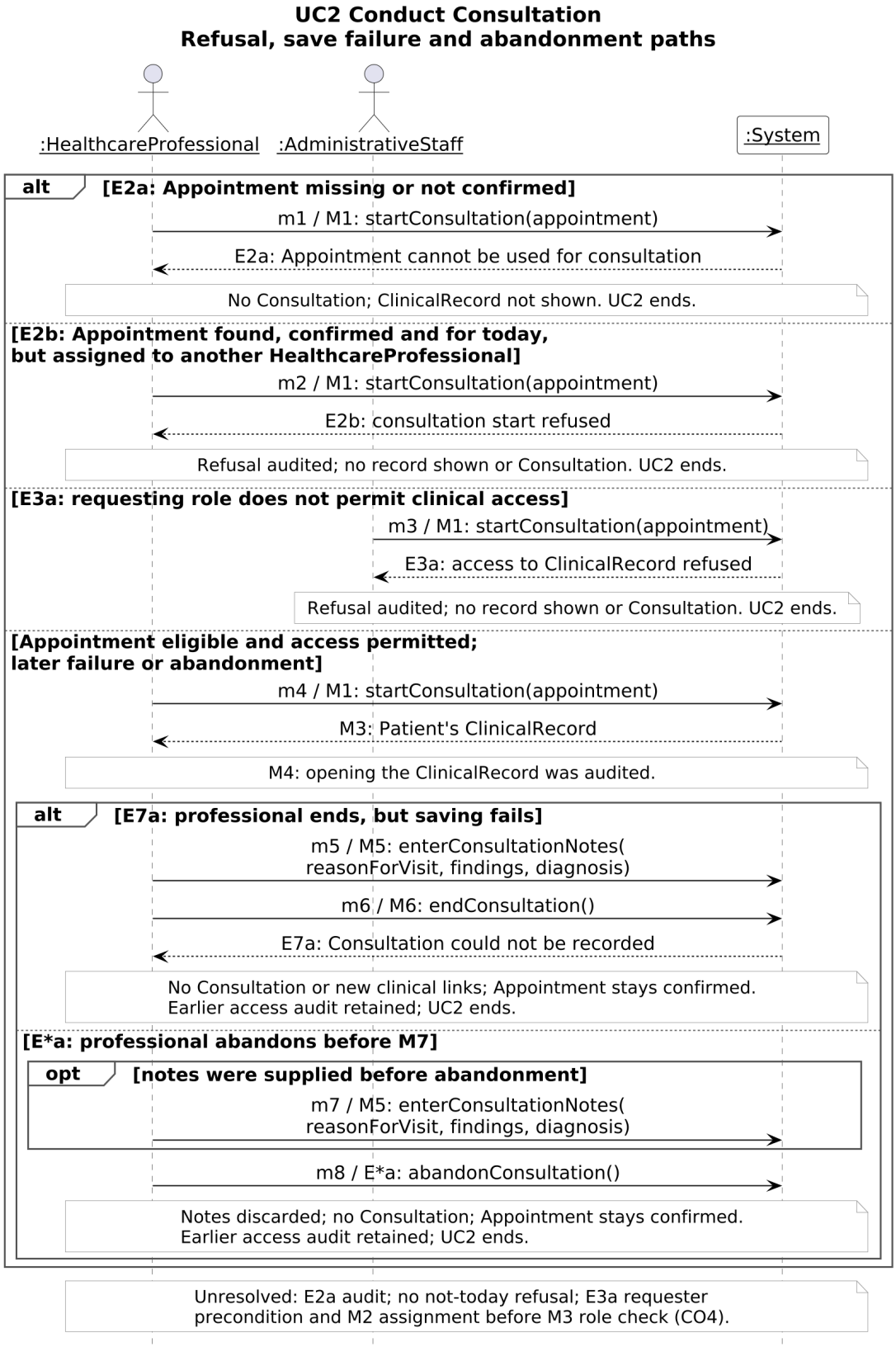
System sequence diagrams — UC2-S

Main steps 1-10, with the documented placement of extensions 5a and 5b. The base success path skips the optional fragment. Each extension returns to step 5 before another extension or completion.



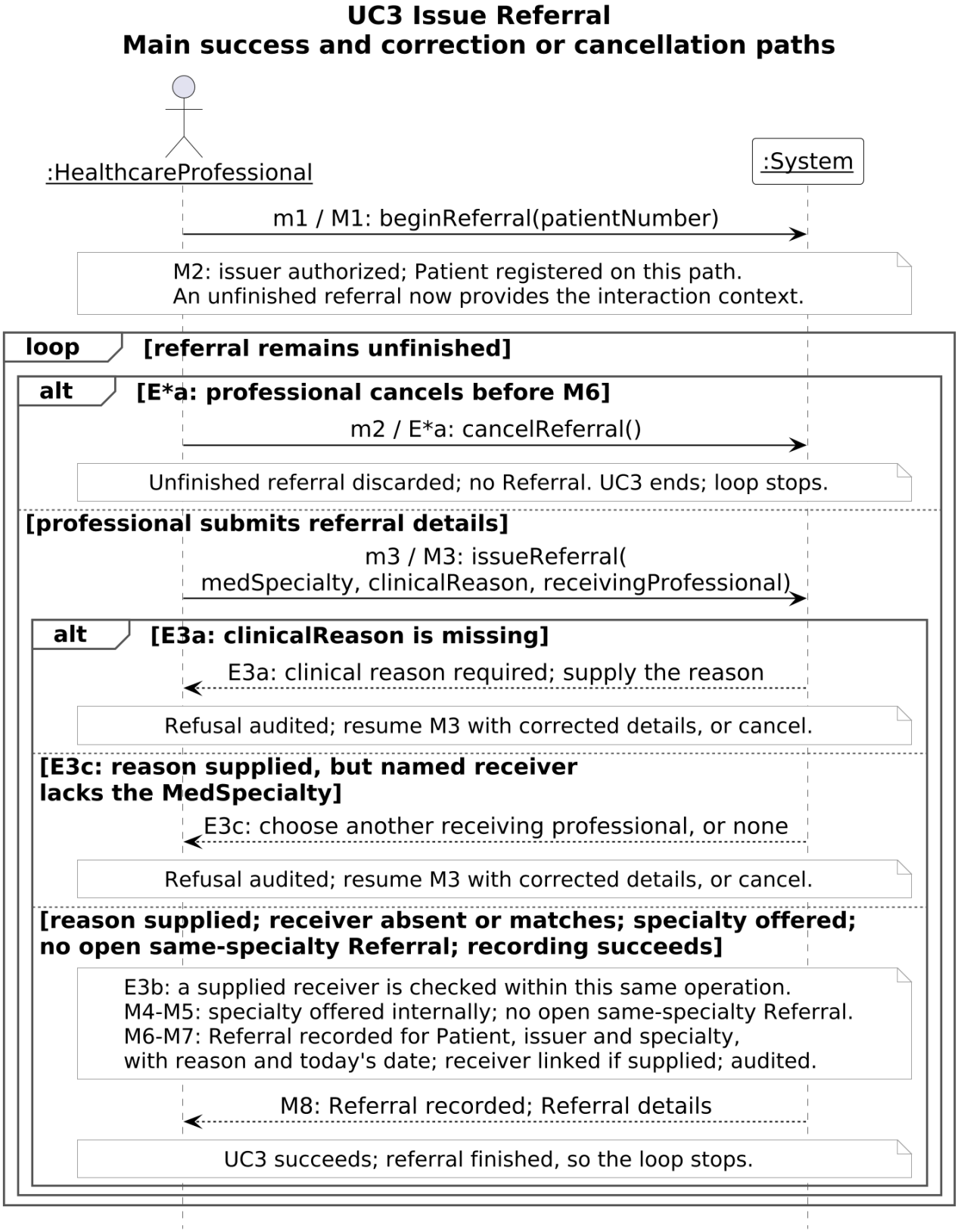
System sequence diagrams — UC2-F

Extensions 2a, 2b, 3a, 7a and *a. Administrative Staff appears only in the documented denied-role branch. Successful access creates an audit that later saving failure or abandonment does not erase.



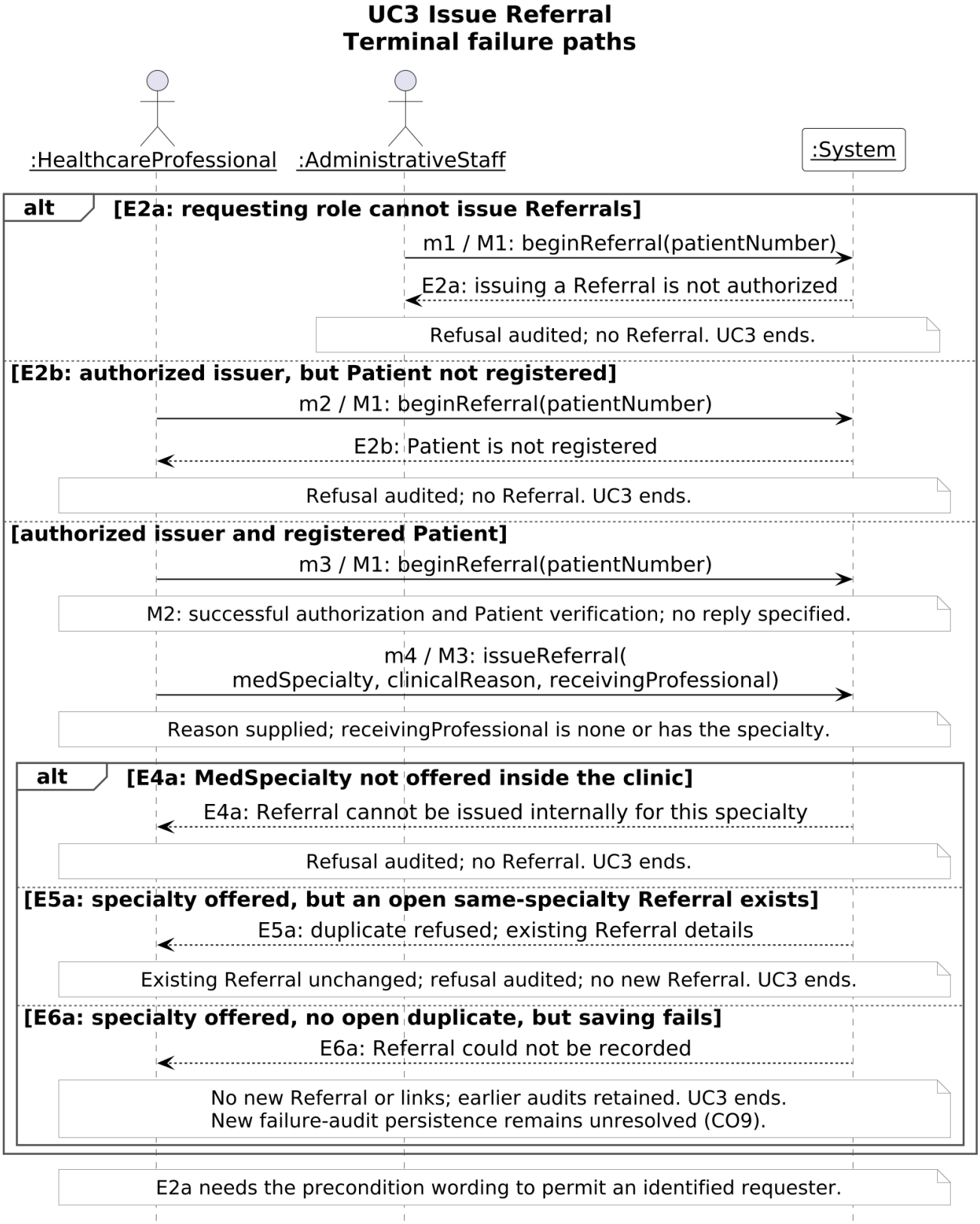
System sequence diagrams — UC3-S

Main steps 1-8; extensions 3a, 3b, 3c and *a. A receiving professional is optional input in issueReferral, not a separate operation. Successful initiation has no specified returned value.



System sequence diagrams — UC3-F

Extensions 2a, 2b, 4a, 5a and 6a. All branches terminate. Duplicate refusal returns the existing Referral unchanged; saving failure does not create a new Referral.



2 System operations

SO01 corresponds to CO1, SO02 to CO2, and so on. M denotes a numbered main-scenario step; E denotes an extension. SSD message identifiers are local to their diagram. State means persistent domain state.

Critical use cases

ID	Operation and parameter types	Origin and SSD evidence	Outcome and persistent state
SO01	beginAppointmentRequest(medicalService: MedicalServiceRef)	UC1 M1-M3; E2a; E3a UC1-S m1; UC1-F m1	Show eligible times. State: refusal audit only.
SO02	selectAppointmentTime(healthcareProfessional: HealthcareProfessionalRef, clinicLocation: ClinicLocationRef, dateTime: DateTime)	UC1 M4-M8; E5a; E5b; E6a UC1-S m3; UC1-F m2	Record Requested Appointment and audit. State: yes.
SO03	cancelAppointmentRequest()	UC1 E*a UC1-S m2	Discard unfinished request. State: no business change.
SO04	startConsultation(appointment: AppointmentRef)	UC2 M1-M4; E2a; E2b; E3a UC2-S m1; UC2-F m1-m4	Validate and open ClinicalRecord. State: access/refusal audit.
SO05	enterConsultationNotes(reasonForVisit: Text, findings: Text, diagnosis: Text)	UC2 M5; E5a/E5b resume M5 UC2-S m2-m3; UC2-F m5, m7	Retain unsaved notes. State: no persistent change.
SO06	endConsultation()	UC2 M6-M10; E7a UC2-S m4; UC2-F m6	Record Consultation; complete Appointment; audit. State: yes.
SO07	abandonConsultation()	UC2 E*a UC2-F m8	Discard unsaved notes. State: no new business change.
SO08	beginReferral(patientNumber: PatientNumber)	UC3 M1-M2; E2a; E2b UC3-S m1; UC3-F m1-m3	Validate issuer/Patient. State: refusal audit only.
SO09	issueReferral(medSpecialty: MedSpecialtyRef, clinicalReason: Text, receivingProfessional: Optional<HealthcareProfessionalRef>)	UC3 M3-M8; E3a; E3b; E3c; E4a; E5a; E6a UC3-S m3; UC3-F m4	Record internal Referral and audit. State: yes.
SO10	cancelReferral()	UC3 E*a UC3-S m2	Discard unfinished referral. State: no business change.

The six SSDs contain 24 incoming message occurrences covering all ten critical operations. Repeated messages reuse the same operation and contract.

Parameter types

XRef is an external selection intended to identify concept X; it may be stale or unresolvable. PatientNumber is Patient.patientNumber. Text, Date, DateTime and Money are values; Optional<T> permits none; Sequence<T> is a finite sequence; DateRange is a start/end time frame. These interface types do not add domain classes.

PatientSearchCriteria contains optional insurance number, birth date, phone and partial first/last names. InsuranceInput contains type, number and provider. ExternalPhysicianInput contains name, licenseNumber and contact. MedicationOrder contains a MedicationRef plus dosage, frequency and duration. ReferralPriority is a supplied priority; allowed values are unspecified.

Operations from noncritical scenarios

These operations are inferred from brief scenarios. Their detailed incoming events still require additional SSD specification; CO23 and CO24 retain material conditions that the source scenarios have not resolved.

ID	Operation and parameter types	Origin and SSD evidence	Outcome and persistent state
SO11	searchPatients(criteria: PatientSearchCriteria)	NC4 S1-S3a; NC1 S2; NC6 S2; NC7 S1a Brief scenario only	Find matching Patients. State: no.
SO12	selectPatient(patientNumber: PatientNumber)	NC4 S3b-S4; NC1 S2-S3a Brief scenario only	Select Patient; show permitted details. State: no.
SO13	findAppointments(patientNumber: PatientNumber)	NC2 S1a Brief scenario only	Find Patient appointments. State: no.
SO14	registerArrival(appointment: AppointmentRef)	NC1 S3b-S4 Brief scenario only	Set arrived and arrivalTime; audit. State: yes.
SO15	confirmAppointment(appointment: AppointmentRef)	NC2 S1b, S4-S5 Brief scenario only	Set confirmed; audit. State: yes.
SO16	rescheduleAppointment(appointment: AppointmentRef, newDateTime: DateTime)	NC2 S1b, S2-S3, S5 Brief scenario only	Validate and change dateTime; audit. State: yes.
SO17	cancelAppointment(appointment: AppointmentRef)	NC2 S1b, S4-S5 Brief scenario only	Set cancelled; audit. State: yes.
SO18	registerPatient(firstName: Text, lastName: Text, dateOfBirth: Date, phone: Text, insurance: Optional<InsuranceInput>)	NC3 S1-S4 Brief scenario only	Create Patient, ClinicalRecord, optional Insurance; audit. State: yes.
SO19	generateInvoice(appointment: AppointmentRef)	NC5 S1-S2 Brief scenario only	Create Invoice from service fee. State: yes.
SO20	recordPayment(invoice: InvoiceRef, amount: Money, method: Text)	NC5 S3-S4 Brief scenario only	Record Payment and invoice status; audit. State: yes.
SO21	recordExternalReferral(patientNumber: PatientNumber, physician: ExternalPhysicianInput, issueDate: Date, clinicalReason: Text, priority: ReferralPriority, requestedSpecialty: Optional<MedSpecialtyRef>, requestedProfessional: Optional<HealthcareProfessionalRef>)	NC6 S2-S4 Brief scenario only	Record external Referral/issuer/targets; audit. State: yes.
SO22	viewClinicalRecord(patientNumber: PatientNumber)	NC7 S1b-S4 Brief scenario only	Authorize and show ClinicalRecord. State: access/refusal audit.
SO23	issuePrescription(consultation: ConsultationRef, items: Sequence<MedicationOrder>, instructions: Optional<Text>)	NC8 S2-S3; UC2 E5a Brief scenario only	Record Prescription and items; audit. State: conditional.
SO24	requestFollowUpAppointment(consultation: ConsultationRef, medicalService: MedicalServiceRef, timeFrame: DateRange)	NC9 S1-S3; UC2 E5b Brief scenario only	Find/book follow-up; audit. State: conditional.

NC1 Register Arrival; NC2 Manage Appointment; NC3 Register Patient; NC4 Search Patient; NC5 Process Uninsured Payment; NC6 Record External Referral; NC7 View Clinical Record; NC8 Issue Prescription; NC9 Request Follow-up Appointment. S references label source sentences/clauses for traceability; the original brief has no numbered steps.

3 Operation contracts

Contracts describe conceptual state changes using the existing domain model. Unrelated instances, attributes and associations remain unchanged. Unsaved input is interaction context, not a new persistent class. Documented rejection conditions are checked during the operation.

Contract CO1: beginAppointmentRequest

Operation: beginAppointmentRequest(medicalService: MedicalServiceRef)

Cross References: UC1 M1-M3; E2a.1; E3a.1. SO01.

Preconditions: Patient p is registered and identified; medicalService denotes existing MedicalService s. Referral validity and availability are checked, not assumed.

Postconditions:

1. **Instances:** Accepted start: none. E2a/E3a refusal: an ActivityRecord ar was created; no Appointment was created.
2. **Attributes:** Accepted start: none. On refusal, ar.timestamp recorded the attempt time, ar.action the refusal reason, and ar.performedBy the identified actor.
3. **Associations:** Accepted start: none. On refusal, ar was associated with p through concerns.

Failures: E2a: a required valid Referral is missing; explain the requirement and end. E3a: no open times; report unavailability and end. Neither failure reserves time or modifies Schedule.

Notes: Acceptance establishes unfinished interaction context and returns eligible ClinicLocations, HealthcareProfessionals, and open Schedule times. Referral validity/applicability and appointment interval rules remain undefined; no persistent request class is introduced.

Contract CO2: selectAppointmentTime

Operation: selectAppointmentTime(healthcareProfessional: HealthcareProfessionalRef, clinicLocation: ClinicLocationRef, dateTime: DateTime)

Cross References: UC1 M4-M8; E5a.1; E5b.1; E6a.1-2. SO02.

Preconditions: Successful SO01 established unfinished context for identified Patient p and MedicalService s. The references select offered HealthcareProfessional h and ClinicLocation l.

Postconditions:

1. **Instances:** Successful save: Appointment a and ActivityRecord ar were created. E5a/E5b rejection: only ar was created. Nothing was deleted.
2. **Attributes:** On success, a.dateTime was set to dateTime and a.status to Requested. On success or E5a/E5b rejection, ar.timestamp, ar.action, and ar.performedBy recorded the attempt time, outcome, and identified actor. No arrival or Schedule attributes changed.
3. **Associations:** On success, p was associated with a through books; h was associated with a through is booked for; a was associated with l through held at and with s through is for. If s.requiresReferral, applicable Referral r was associated with a through results in. On success or E5a/E5b rejection, ar was associated with p through concerns.

Failures: E5a: time taken; reject and refresh options, returning to M3. E5b: Patient overlap; identify p's conflict and return to M3. Both create no Appointment/reservation. E6a: save failure; report and end, recording no Appointment/links and leaving the time open. Earlier audits remain; this failure's audit is unresolved because "nothing changed" conflicts with mandatory refusal logging.

Notes: Availability and Patient overlap are validated. Success returns details and Requested status pending staff confirmation. The Appointment holds the time; Schedule working hours remain unchanged. Exact interval rules are unspecified.

Contract CO3: cancelAppointmentRequest

Operation: cancelAppointmentRequest()

Cross References: UC1 E*a.1, before M6. SO03.

Preconditions: The identified Patient has an unfinished UC1 request whose Appointment has not been successfully recorded.

Postconditions:

1. **Instances:** No persistent Appointment or other business instance was created or deleted.
2. **Attributes:** None in the documented extension; existing Appointments and Schedule working windows were unchanged.
3. **Associations:** None were formed or broken. Previously saved refusal ActivityRecords and their associations were retained.

Failures: No failure branch is documented for cancellation satisfying these preconditions. Cancellation after recording is outside this operation's scope.

Notes: The unfinished request was discarded and UC1 ended without recording an Appointment. Cancellation auditing is unspecified. Cancelling an existing recorded Appointment is a different operation, SO17.

Contract CO4: startConsultation

Operation: startConsultation(appointment: AppointmentRef)

Cross References: UC2 M1-M4; E2a.1; E2b.1; E3a.1. SO04.

Preconditions: Proposed precondition clarification: the requesting user is identified. The Patient is present at the ClinicLocation for the intended consultation. Appointment eligibility and clinical authorization are validated.

Postconditions:

1. **Instances:** Allowed access or E2b/E3a refusal: an ActivityRecord ar was created. No Consultation was created; nothing was deleted.
2. **Attributes:** For those outcomes, ar.timestamp recorded the attempt time, ar.action the access/refusal outcome, and ar.performedBy the identified actor. Appointment and clinical attributes were unchanged.
3. **Associations:** ar was associated through concerns with Appointment Patient p when known. No clinical association changed.

Failures: E2a: missing/not-confirmed Appointment; refuse/end with the literal no-change guarantee. Whether this also requires a refused-access audit needs clarification. E2b: wrong professional; refuse, audit, and end. E3a: disallowed role; refuse, audit, and end. No rejected start discloses the ClinicalRecord or creates a Consultation.

Notes: Success requires confirmed status, today's date, assignment to current HealthcareProfessional h, and clinical authorization; it returns p's ClinicalRecord and establishes unfinished consultation context. Register Arrival sets arrived, which UC2 rejects. Accepting both statuses and preserving entry status on failure/abandonment is proposed. The source lacks a not-today rejection/audit outcome.

Contract CO5: enterConsultationNotes

Operation: enterConsultationNotes(reasonForVisit: Text, findings: Text, diagnosis: Text)
Cross References: UC2 M5; E5a and E5b returning to M5. SO05.

Preconditions: SO04 established unfinished consultation context for the authorized HealthcareProfessional, Appointment a, and Patient p. The consultation has not ended or been abandoned.

Postconditions:

1. **Instances:** None in persistent domain state.
2. **Attributes:** None in persistent domain state.
3. **Associations:** None in persistent domain state.

Failures: No note-validation failure is documented; no required-field rule is introduced.

Notes: Submitted reason, findings, and diagnosis became unsaved notes available to SO06. Proposed mapping: retain reason/findings within existing Consultation.observations when recording. No new notes attribute is assumed. Repeated submissions' append/replace behavior is unspecified. Existing treatmentRecommendations and followUpRequired attributes have no supplied actor inputs.

Contract CO6: endConsultation

Operation: endConsultation()
Cross References: UC2 M6-M10; E7a.1-2; base success excludes E5a/E5b. SO06.

Preconditions: SO04 established context for Appointment a, Patient p, and the authorized professional; SO05 supplied notes. The professional has neither abandoned nor finalized it; no persistent Consultation exists for this invocation. Successful saving is not a precondition.

Postconditions:

1. **Instances:** Successful save: Consultation c and ActivityRecord ar were created. Nothing was deleted.
2. **Attributes:** c.diagnosis was set to the submitted diagnosis. Proposed mapping: c.observations retained the submitted reason/findings. a.status was set to completed. ar.timestamp, ar.action, and ar.performedBy recorded the recording time, successful consultation, and identified professional. Values for treatmentRecommendations/followUpRequired remain unspecified.
3. **Associations:** a was associated with c through results in; existing ClinicalRecord cr was associated with c through contains; ar was associated with Patient p through concerns. No association was broken.

Failures: E7a: report save failure and end. No Consultation/new links were recorded; a.status remained confirmed in the unchanged source. No success audit was created; the earlier ClinicalRecord-access audit remains. Proposed eligibility correction: preserve the confirmed/arrived entry status. Persistent draft retention is unspecified.

Notes: Success confirms recording and ends UC2. These are base-flow postconditions. Optional prescriptions/follow-up (CO23/CO24) currently link to a Consultation before its creation. Staging their inputs and committing with c is proposed; extension completion remains unresolved.

Contract CO7: abandonConsultation

Operation: abandonConsultation()

Cross References: UC2 E*a.1-2, before M7. SO07.

Preconditions: An unfinished consultation exists in interaction context for the identified professional and Appointment a. SO04 succeeded; no Consultation has been recorded.

Postconditions:

1. **Instances:** No persistent Consultation or other business instance was created or deleted.
2. **Attributes:** a.status remained confirmed under the unchanged extension. No clinical content attribute changed.
3. **Associations:** None changed. The earlier ClinicalRecord-access ActivityRecord and its Patient association were retained.

Failures: No separate failure branch is documented. This operation does not remove an already recorded Consultation.

Notes: Unsaved notes were discarded and UC2 ended. No new abandonment audit is specified. Proposed eligibility correction: preserve the confirmed/arrived entry status. Optional extensions' products need a persistence rule; successful independent recording cannot be assumed.

Contract CO8: beginReferral

Operation: beginReferral(patientNumber: PatientNumber)

Cross References: UC3 M1-M2; E2a.1; E2b.1. SO08.

Preconditions: Proposed precondition clarification: the requesting user is identified. Authorization and existence of Patient.patientNumber = patientNumber are checked, not assumed.

Postconditions:

1. **Instances:** Accepted start: none. E2a/E2b refusal: an ActivityRecord ar was created; no Patient or Referral was created.
2. **Attributes:** Accepted start: none. On refusal, ar.timestamp recorded attempt time, ar.action the refusal reason, and ar.performedBy the identified actor.
3. **Associations:** Accepted start: none. On refusal, ar was associated with the resolved Patient through concerns; if none was resolved, the optional Patient endpoint remained absent.

Failures: E2a: unauthorized role; refuse and end. E2b: Patient absent; report that registration is required and end. No clinical information is disclosed to a disallowed role.

Notes: Acceptance establishes unfinished internal-referral context for authorized HealthcareProfessional h and Patient p; no Referral is yet recorded. Acknowledgement may invite referral details. A missing patient's submitted number can appear in action text without adding an audit attribute.

Contract CO9: issueReferral

Operation: issueReferral(medSpecialty: MedSpecialtyRef, clinicalReason: Text, receivingProfessional: Optional<HealthcareProfessionalRef>)

Cross References: UC3 M3-M8; E3a.1; E3b.1; E3c.1; E4a.1; E5a.1; E6a.1. SO09.

Preconditions: SO08 established context for registered Patient p and authorized issuing HealthcareProfessional h. References denote existing MedSpecialty s and, if supplied, HealthcareProfessional receiver.

Postconditions:

1. **Instances:** Success: Referral r and ActivityRecord ar were created. E3a/E3c/E4a/E5a rejection: only ar was created. Nothing was deleted.
2. **Attributes:** Success: r.clinicalReason was set to clinicalReason; r.issueDate to today. Initial priority/status are unspecified. On success or listed rejection, ar.timestamp, ar.action, and ar.performedBy recorded attempt time, outcome, and identified actor.
3. **Associations:** On success, r was associated with p through concerns; h was associated with r through refers; r was associated with s through requests specialty and, optionally, with receiver through

requests professional. No ExternalPhysician issuer was linked. On success or listed rejection, ar was associated with p through concerns.

Failures: E3a: missing reason; return to M3. E3c: receiver lacks specialty; request another or none, returning to M3. E4a: specialty unavailable; end. E5a: open duplicate; present existing unchanged Referral and end. These create no Referral. E6a: save failed; report/end without new Referral/links; earlier audits remain. This failure's audit is unresolved: "nothing changed" conflicts with mandatory refusal logging.

Notes: Reason, receiver match, specialty availability, and no open duplicate are validated. Availability requires a HealthcareProfessional with s practising at a ClinicLocation. A matching optional receiver continues at M4 (E3b). Success presents r without booking. The meaning of open remains undefined.

Contract CO10: cancelReferral

Operation: cancelReferral()

Cross References: UC3 E*a.1, before M6. SO10.

Preconditions: The requesting HealthcareProfessional and Patient have unfinished UC3 internal-referral context; no Referral has been successfully recorded for it.

Postconditions:

1. **Instances:** No persistent Referral or other business instance was created or deleted.
2. **Attributes:** None in the documented cancellation extension. No existing Referral was assigned a cancelled status.
3. **Associations:** None were formed or broken. Previously saved refusal ActivityRecords and their associations remained.

Failures: No failure branch is documented for cancellation satisfying these preconditions. A previously issued Referral cannot be revoked by this operation.

Notes: The unfinished referral was discarded and UC3 ended without recording it. Voluntary-cancellation auditing is unspecified. Cancelling an unfinished request does not undo earlier successful domain changes.

Contract CO11: searchPatients

Operation: searchPatients(criteria: PatientSearchCriteria)

Cross References: NC4 S1-S3a; NC1 S2; NC6 S2; NC7 S1a

Preconditions: The identified actor is Administrative Staff or a Healthcare Professional performing a patient search. No matching Patient is assumed to exist.

Postconditions:

1. **Instances:** No persistent instance was created or deleted; administrative search does not expressly require an ActivityRecord.
2. **Attributes:** No Patient, Insurance, or other persistent attribute was changed.
3. **Associations:** No persistent association was formed or broken.

Failures: Empty criteria, no matches, invalid criteria, and retrieval-failure responses are unspecified.

Notes: Returns role-permitted matches using the supplied insurance number, birth date, phone, or partial names. Matching/combination rules need clarification. Clinical-record disclosure is not included in this administrative search contract.

Contract CO12: selectPatient

Operation: selectPatient(patientNumber: PatientNumber)
Cross References: NC4 S3b-S4; NC1 S2-S3a

Preconditions: The identified staff member or Healthcare Professional selects Patient p from preceding search results using p.patientNumber.

Postconditions:

1. **Instances:** No persistent instance was created or deleted by administrative selection.
2. **Attributes:** No persistent attribute was changed.
3. **Associations:** No persistent association was formed or broken; the selected patient is interaction context.

Failures: Unknown or stale selections have no specified failure response.

Notes: Returns permitted patient details; Administrative Staff receive administrative information. In Register Arrival, the response includes today's appointments. Any intended clinical disclosure on selection needs explicit authorization and auditing; this contract does not silently treat clinical access as an unlogged read.

Contract CO13: findAppointments

Operation: findAppointments(patientNumber: PatientNumber)
Cross References: NC2 S1a

Preconditions: An identified Administrative Staff member is managing appointments for the existing Patient p identified by patientNumber.

Postconditions:

1. **Instances:** No persistent instance was created or deleted.
2. **Attributes:** No persistent attribute was changed.
3. **Associations:** No persistent association was formed or broken.

Failures: Unknown patients, empty results, and retrieval failures are unspecified.

Notes: Returns administrative information about p's existing Appointments through Patient - books - Appointment. This lookup is inferred from NC2's "finds an existing appointment"; its separate incoming event still needs an expanded scenario/SSD. No date filter, confirmation, cancellation, or lookup audit is added.

Contract CO14: registerArrival

Operation: registerArrival(appointment: AppointmentRef)
Cross References: NC1 S3b-S4

Preconditions: An identified Administrative Staff member registers Patient p's arrival and selects existing Appointment a from p's appointments for today.

Postconditions:

1. **Instances:** ActivityRecord ar was created; a and p were retained.
2. **Attributes:** a.arrivalTime was set to arrival-registration time t; a.status to arrived. ar.timestamp was set to t, ar.action described arrival registration, and ar.performedBy identified the actor.
3. **Associations:** ar was associated with p through Patient - concerns - ActivityRecord. Existing Appointment associations were retained.

Failures: Repeated arrival, invalid prior status, and saving-failure behavior are unspecified.

Notes: The source's arrived status conflicts with UC2's confirmed-only eligibility. Resolve that shared lifecycle rule without silently changing this documented arrival outcome.

Contract CO15: confirmAppointment

Operation: confirmAppointment(appointment: AppointmentRef)
Cross References: NC2 S1b, S4-S5

Preconditions: An identified Administrative Staff member has found existing Appointment a for Patient p and chosen confirmation.

Postconditions:

1. **Instances:** ActivityRecord ar was created; a was retained.
2. **Attributes:** a.status was set to confirmed. ar.timestamp recorded the event time, ar.action described appointment confirmation, and ar.performedBy identified the actor.
3. **Associations:** ar was associated with p through Patient - concerns - ActivityRecord. Existing Appointment associations were retained.

Failures: Permitted prior statuses, repeat confirmation, and saving-failure outcomes are unspecified.

Notes: Confirmation saves the status change. A Requested-only precondition is not established by the brief. No second reservation or Schedule attribute change is introduced; UC1 already states that a Requested appointment holds its time.

Contract CO16: rescheduleAppointment

Operation: rescheduleAppointment(appointment: AppointmentRef, newDateTime: DateTime)
Cross References: NC2 S1b, S2-S3, S5

Preconditions: An identified Administrative Staff member has selected existing Appointment a for Patient p, retaining its HealthcareProfessional h and ClinicLocation l.

Postconditions:

1. **Instances:** On accepted rescheduling, ActivityRecord ar was created; a retained its identity.
2. **Attributes:** a.dateTime was set to newDateTime. ar.timestamp recorded the event time, ar.action described rescheduling, and ar.performedBy identified the actor.
3. **Associations:** ar was associated with p through Patient - concerns - ActivityRecord. a retained its existing Patient, HealthcareProfessional, ClinicLocation, and MedicalService associations.

Failures: The brief does not specify an availability-rejection branch. Proposed rejection guarantee: leave a.dateTime unchanged when the requested time fails validation. The refusal response, retry, failure auditing and storage-failure guarantees require specification.

Notes: No rescheduling status assignment is supplied. Interval duration remains undefined. Booking occupancy is derived from appointment facts; Schedule working hours are not changed.

Contract CO17: cancelAppointment

Operation: cancelAppointment(appointment: AppointmentRef)
Cross References: NC2 S1b, S4-S5

Preconditions: An identified Administrative Staff member has found existing Appointment a for Patient p and chosen cancellation.

Postconditions:

1. **Instances:** ActivityRecord ar was created; a was retained.
2. **Attributes:** a.status was set to cancelled. ar.timestamp recorded the event time, ar.action described cancellation, and ar.performedBy identified the actor.
3. **Associations:** ar was associated with p through Patient - concerns - ActivityRecord. Existing Appointment associations, including any Referral link, were retained.

Failures: Permitted prior statuses, repeat cancellation, and saving-failure outcomes are unspecified.

Notes: This changes a stored appointment's status. UC1's cancelAppointmentRequest discards unfinished input instead. The shared availability rule must specify released occupancy; no Appointment deletion, Referral consumption, refund, or independent slot-release operation is asserted.

Contract CO18: registerPatient

Operation: registerPatient(firstName: Text, lastName: Text, dateOfBirth: Date, phone: Text, insurance: Optional<InsuranceInput>)

Cross References: NC3 S1-S4

Preconditions: An identified Administrative Staff member is registering a patient. Absence of an existing registration is checked during the operation.

Postconditions:

1. **Instances:** On acceptance, Patient p, empty ClinicalRecord cr, and ActivityRecord ar were created; Insurance i was created if supplied.
2. **Attributes:** p.patientNumber received a unique value; p's firstName, lastName, dateOfBirth, and phone received the inputs. If supplied, i.type, i.number, and i.provider received insurance values. ar.timestamp recorded the event time, ar.action described registration, and ar.performedBy identified the actor.
3. **Associations:** p was associated with cr through Patient - has - ClinicalRecord and, when supplied, with i through Patient - has - Insurance. ar was associated with p through concerns. No Consultation association was formed.

Failures: Duplicate matching, rejection effects, and saving failures are unspecified.

Notes: ClinicalRecord.recordNumber/createdDate initialization needs agreement. No insurer verification or coverage determination occurs.

Contract CO19: generateInvoice

Operation: generateInvoice(appointment: AppointmentRef)

Cross References: NC5 S1-S2

Preconditions: An identified Administrative Staff member selects Appointment a for an uninsured MedicalService s. This first-invoice branch assumes a has no Invoice, respecting its 0..1 multiplicity.

Postconditions:

1. **Instances:** Invoice inv was created. No Payment was created; invoice-generation auditing is not separately specified.
2. **Attributes:** inv.amount was set to the applicable s.fee.
3. **Associations:** a was associated with inv through Appointment - billed by - Invoice. No Payment association was formed.

Failures: Repeat generation, missing fees, and saving failures are unspecified.

Notes: Invoice number, date, and initial status need initialization rules. Uninsured-service eligibility is scenario context; lack of Patient insurance alone does not establish it. No external claims or payment-processing interaction is implied.

Contract CO20: recordPayment

Operation: recordPayment(invoice: InvoiceRef, amount: Money, method: Text)

Cross References: NC5 S3-S4

Preconditions: An identified Administrative Staff member records payment against existing Invoice inv for Appointment a and Patient p.

Postconditions:

1. **Instances:** On an accepted payment, Payment pay and ActivityRecord ar were created.
2. **Attributes:** pay.amount and pay.method received the inputs. inv.status became paid if the associated payment total equals inv.amount, or partially paid if positive and smaller. ar.timestamp recorded the event time, ar.action described payment registration, and ar.performedBy identified the actor.
3. **Associations:** inv was associated with pay through Invoice - settled by - Payment. ar was associated with p through Patient - concerns - ActivityRecord. Existing payment associations were retained.

Failures: Zero/negative amounts, overpayments, settled invoices, and saving failures are unspecified.

Notes: Payment.paymentDate initialization needs agreement. This records payment facts; it does not execute an external financial transaction.

Contract CO21: recordExternalReferral

Operation: recordExternalReferral(patientNumber: PatientNumber, physician: ExternalPhysicianInput, issueDate: Date, clinicalReason: Text, priority: ReferralPriority, requestedSpecialty: Optional<MedSpecialtyRef>, requestedProfessional: Optional<HealthcareProfessionalRef>)

Cross References: NC6 S2-S4

Preconditions: An identified Administrative Staff member has found registered Patient p and is recording the patient's externally issued referral.

Postconditions:

1. **Instances:** For an accepted recording with issuer identity resolved, Referral r and ActivityRecord ar were created. ExternalPhysician e was created if previously unrecorded; otherwise the resolved existing e was reused.
2. **Attributes:** r.issueDate, r.clinicalReason, and r.priority received the inputs; r.status became pending. New e received physician.name, licenseNumber, and contact; existing e was not overwritten. ar.timestamp recorded the event time, ar.action described external-referral recording, and ar.performedBy identified the actor.
3. **Associations:** p was associated with r through concerns; e was associated with r through refers. If supplied, r was associated with requestedSpecialty through requests specialty and/or with requestedProfessional through requests professional. ar was associated with p through concerns. No internal issuer association was formed.

Failures: Duplicate external referrals, unresolved physician identity, invalid targets, and saving-failure behavior are unspecified; UC3's duplicate-open rule is not automatically imported.

Notes: The model requires exactly one issuer and at least one requested target. Physician matching and Administrative Staff's clinical-reason capture/read permissions require clarification. Recording the Referral creates no Appointment and asserts no expiry or single-use rule.

Contract CO22: viewClinicalRecord

Operation: viewClinicalRecord(patientNumber: PatientNumber)

Cross References: NC7 S1b-S4; UC2 special requirement on permitted/refused ClinicalRecord access

Preconditions: The identified actor has found Patient p, who has ClinicalRecord cr. Clinical authorization is checked during the operation.

Postconditions:

1. **Instances:** For permitted access, ActivityRecord ar was created; clinical objects were retained.
2. **Attributes:** ar.timestamp recorded the access time, ar.action described opening cr, and ar.performedBy identified the actor. Clinical content remained unchanged.
3. **Associations:** ar was associated with p through Patient - concerns - ActivityRecord. Existing clinical associations were retained.

Failures: On denied access, no clinical content was returned. Under UC2's special requirement, a denial ActivityRecord was created, its timestamp/action/performedBy recorded the attempt time, refusal, and actor, and it was associated with known Patient p through concerns. Unknown-patient and audit-failure handling are unspecified.

Notes: Returns permitted Consultations and Prescriptions. UC2 already opens its record within startConsultation; no extra actor event is added there.

Contract CO23: issuePrescription

Operation: issuePrescription(consultation: ConsultationRef, items: Sequence<MedicationOrder>, instructions: Optional<Text>)

Cross References: NC8 S2-S3; UC2 E5a

Preconditions: The actor is the Healthcare Professional conducting the consultation for Patient p. At an agreed successful persistence boundary, Consultation c must exist or be created in that same transition. Unchanged UC2 E5a does not yet satisfy this condition.

Postconditions:

1. **Instances:** Conditionally, Prescription rx and ActivityRecord ar were created. Each selected Medication received a PrescriptionItem association-class instance pi for its includes link. Existing Medication instances were reused.
2. **Attributes:** rx.issueDate received the successful issue date; supplied instructions populated rx.instructions. Each pi.dosage, pi.frequency, and pi.duration received its order's values. ar.timestamp recorded issuance time, ar.action described issuance, and ar.performedBy identified the actor.
3. **Associations:** c was associated with rx through issues. rx was associated with each selected Medication through includes, with pi qualifying that existing association. ar was associated with p through concerns. No separate PrescriptionItem endpoint associations were formed.

Failures: The brief specifies no save failure, cancellation, or duplicate-medication policy; no rollback guarantee is inferred.

Notes: Conditional end-state only until consultation/extension timing is resolved. One or more medications are required. Staging inputs and persisting at UC2 completion is a proposal, not the existing flow. Pharmacy transmission is excluded.

Contract CO24: requestFollowUpAppointment

Operation: requestFollowUpAppointment(consultation: ConsultationRef, medicalService: MedicalServiceRef, timeFrame: DateRange)

Cross References: NC9 S1-S3; UC2 E5b

Preconditions: The actor is the Healthcare Professional h conducting the consultation for Patient p. At an agreed successful persistence boundary, Consultation c must exist or be created simultaneously; unchanged UC2 E5b precedes that boundary.

Postconditions:

1. **Instances:** Conditionally on a defined successful booking, follow-up Appointment f and ActivityRecord ar were created.
2. **Attributes:** Under an agreed selection/lifecycle rule, f.dateTime received chosenTime within timeFrame and f.status received chosenStatus. Both remain unresolved values. ar.timestamp recorded booking time, ar.action described follow-up booking, and ar.performedBy identified the actor. No unconditional c.followUpRequired assignment is asserted.
3. **Associations:** p was associated with f through books; h was associated with f through is booked for. f was associated with medicalService through is for and with chosenLocation through held at. c was associated with f through Requests follow-up. ar was associated with p through concerns. The original consultation/appointment association was retained.

Failures: Unavailable times, concurrent slot loss, cancellation, and saving-failure outcomes are unspecified.

Notes: Conditional end-state only. The brief supplies a service/time frame, finds open times, and states booking, but provides no exact-time/location selection or initial status. chosenTime/chosenLocation/chosenStatus are unresolved symbols, not new inputs or automatic choices. Consultation timing, interval data, and applicable referral/scheduling conditions need clarification.

Sources: main branch a67b89c32ec43a17f5c4a385fff68f8f013d3a95; UC1, UC2 and UC3 documents; non_critical_usecases.md; DomainModel.drawio.png; p01.iter1.pdf, pp. 2-4. Method: Lecture 7, pp. 2-7; Lecture 8, pp. 2-10; Lecture 9, p. 7; Lecture 10, pp. 6-7 and 15-17; Lecture 11, p. 8 (instance notation). Proposed clarifications remain explicitly identified in the relevant contracts.